10/01/25 M/S

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# FIRST SEMESTER B.Tech EXAMINATION, JANUARY– 2025

## COMPUTATIONAL CHEMISTRY

**Time: 3 Hours** **Maximum Marks: 100**

**Instructions:** i. Missing data may be suitably assumed.

**ANSWER ALL QUESTIONS**

| | | | Blooms Level | CO | Marks |
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| | | **PART-A** | **5 X 2=10** | | |
| **1.** | **a.** | Mention any two applications of computational chemistry. | L-1 | CO1 | **02** |
| | **b.** | Differentiate between a closed and an isolated system. | L-2 | CO2 | **02** |
| | **c.** | Define single electrode potential and standard electrode potential. | L-1 | CO3 | **02** |
| | **d.** | Describe the orientational order and positional order of liquid crystals. | L-2 | CO4 | **02** |
| | **e.** | Mention the different types of hard disks. | L-1 | CO5 | **02** |
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| | | **PART-B** | **6 X 5=30** | | |
| **2.** | **a.** | Distinguish between covalent and ionic bonds with the help of examples. | L-2 | CO1 | **05** |
| | | **OR** | | | |
| | **b.** | How is a theory different from a model? Explain with the help of examples. | L-1 | CO1 | **05** |
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| **3.** | **a.** | Explain intensive and extensive properties of a system. Give two examples for each. | L-2 | CO2 | **05** |
| | | **OR** | | | |
| | **b.** | Derive the expression for partial molar free energy. | L-2 | CO2 | **05** |
| | | | | | |
| **4.** | **a.** | Differentiate between primary and secondary batteries. Provide examples for both. | L-2 | CO3 | **05** |
| | | **OR** | | | |
| | **b.** | Explain the construction and working principle of Daniel cell. | L-1 | CO3 | **05** |
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| **5.** | **a.** | Illustrate the construction of a liquid crystal display with the help of a neat diagram. | L-2 | CO4 | **05** |
| | | | | | |
| | **b.** | What is meant by carbon nanotubes? Explain the types and properties of carbon nanotubes. | L-2 | CO4 | **05** |

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| **6.** | **a.** | What is meant by e-waste? Mention some health and environmental effects caused by e-wastes. | L-1 | CO5 | **05** |
| | | **OR** | | | |
| | **b.** | Outline the properties and application of nickel in computer hardware. | L-2 | CO5 | **05** |
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| **7.** | **a.** | What is meant by a graphene polymer composite? Mention any two graphene polymer composites and their application. | L-1 | CO4 | **05** |
| | | **OR** | | | |
| | **b.** | List out the applications of liquid crystals. | L-2 | CO4 | **05** |
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| | | **PART-C** | **6 X 10=60** | | |
| **8.** | **a.** | What is meant by the Z-matrix representation of a molecule? How many variables are required to represent a linear and a non-linear molecule (each having N atoms) using the Z-matrix representation?<br><br>Consider a hypothetical molecule ABC with bonding patterns as given below.<br><br>A — B<br>\|<br>C<br><br>The bond lengths of AB and BC bond 1 Å and 1.2 Å, respectively. The A-B-C bond angle is 90°. Write down the Z-matrix for this molecule. | L-2 | CO1 | **10** |
| | | **OR** | | | |
| | **b.** | Briefly explain any four models used in computational chemistry. | L-2 | CO1 | **10** |
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| **9.** | **a.** | What is meant by heat transfer? Explain the three modes of heat transfer. | L-1 | CO2 | **10** |
| | | **OR** | | | |
| | **b.** | Illustrate entropy with the help of any three equations. Mention the properties of entropy. | L-3 | CO2 | **10** |
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| **10.** | **a.** | Explain the structure and working of lithium ion batteries. Mention some of their advantages and disadvantages. | L-2 | CO3 | **10** |

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| | | **OR** | | | |
| | **b.** | Explain the working of a supercapacitor with the help of a neat diagram. Mention an electrode, an electrolyte and a separator material used in supercapacitors. | L-2 | CO3 | **10** |
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| **11.** | **a.** | Explain thermotropic liquid crystals. Illustrate smectic, nematic and ferroelectric liquid crystals. | L-1 | CO4 | **10** |
| | | **OR** | | | |
| | **b.** | Discuss the various steps involved in the sol-gel method for the synthesis of nano materials. | L-1 | CO4 | **10** |
| | | | | | |
| **12.** | **a.** | What is meant by e-waste management? Briefly explain any four e-waste minimization techniques. | L-1 | CO5 | **10** |
| | | **OR** | | | |
| | **b.** | What are permanent magnets? Explain any four types of permanent magnets. | L-1 | CO5 | **10** |
| | | | | | |
| **13.** | **a.** | What is meant by energy storage? Explain the necessity and the general benefits of energy storage. | L-2 | CO3 | **10** |
| | | **OR** | | | |
| | **b.** | What are magnetic materials? How are magnetic materials classified based on the direction and magnitude of magnetization? | L-2 | CO5 | **10** |